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A

TODO

* Check courtyards of components.

* Diode footprints/model

MAYBE

* Upgrade OT1-6 Dual MOSFETs

* Connector for front-panel USB socket - should probably go to raspberry-pi - need a small break-out PCB in that case.

DONE

* 2xI2C mux SDA/SDC/INT signals on a header

* Used 8xI2C mux INT signals.

* Load-cell amplifier (HX717)

* VUSB sense input.

* Additional 5V rail for onboard and up-lights.

* Raspberry Pi 5 power connector

* Supercap/RTC osc - deleted.

* Models for JST-XH (e.g 'B4B-XH-A(LF)(SN)')

* Ethernet PHY

* MOSFET + connector for external ethernet link leds.

* 3V3 power from core board

* LEDs for power rails.

* RGB LED output, for up-lights

* 2x vacuum sensors.

* Head RS422 (non-isolated power).

* Analog mux (2x1 in 4 out).

NO

* reverse polarity protection, need multiple high-current fets like SUD50P06-15-GE3.

B

C

D

E

F

G

H

Itemref

Quantity

Title/Name, designation, material, dimension etc

Article No./Reference

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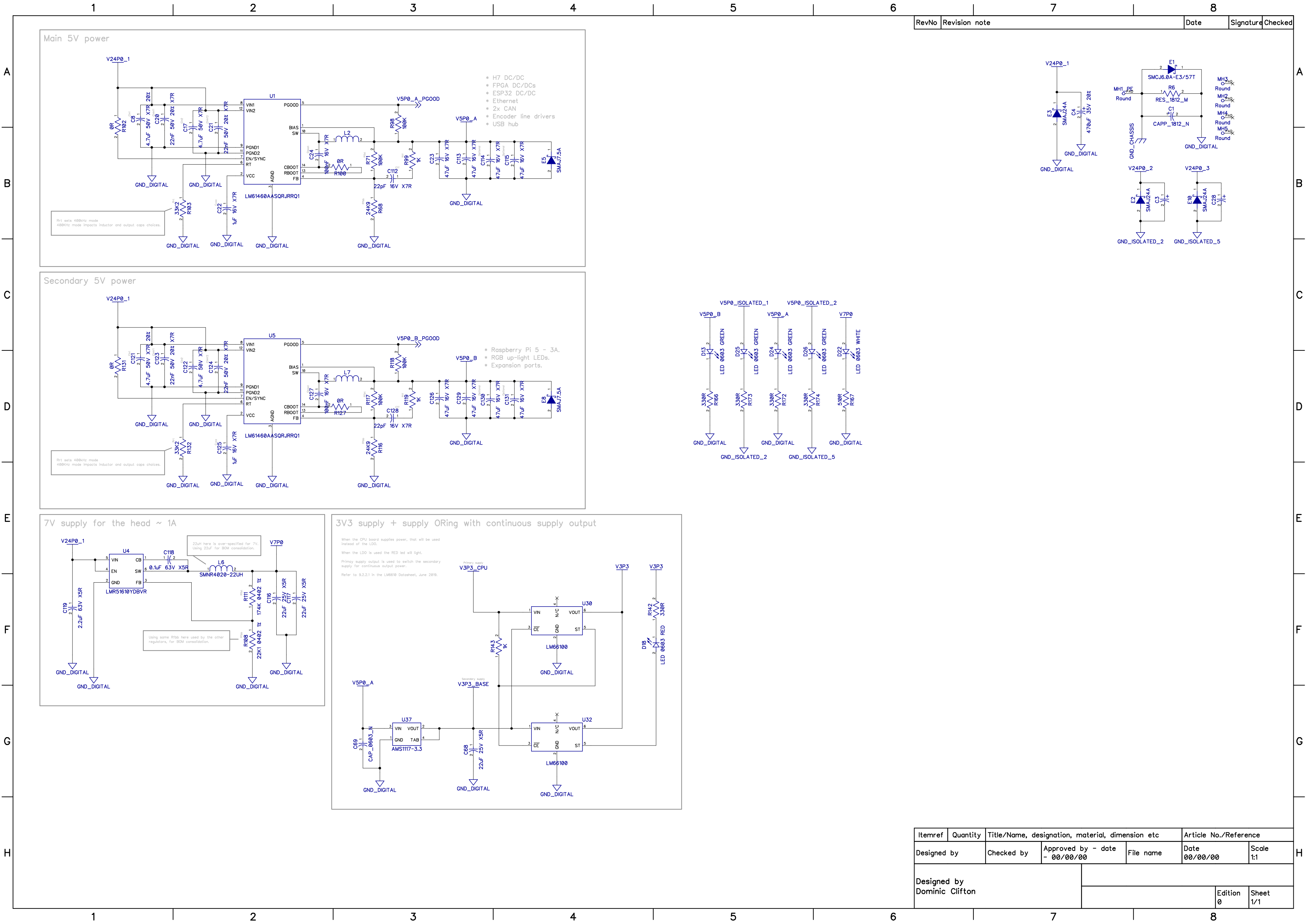
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Designed by
Dominic Clifton

Edition
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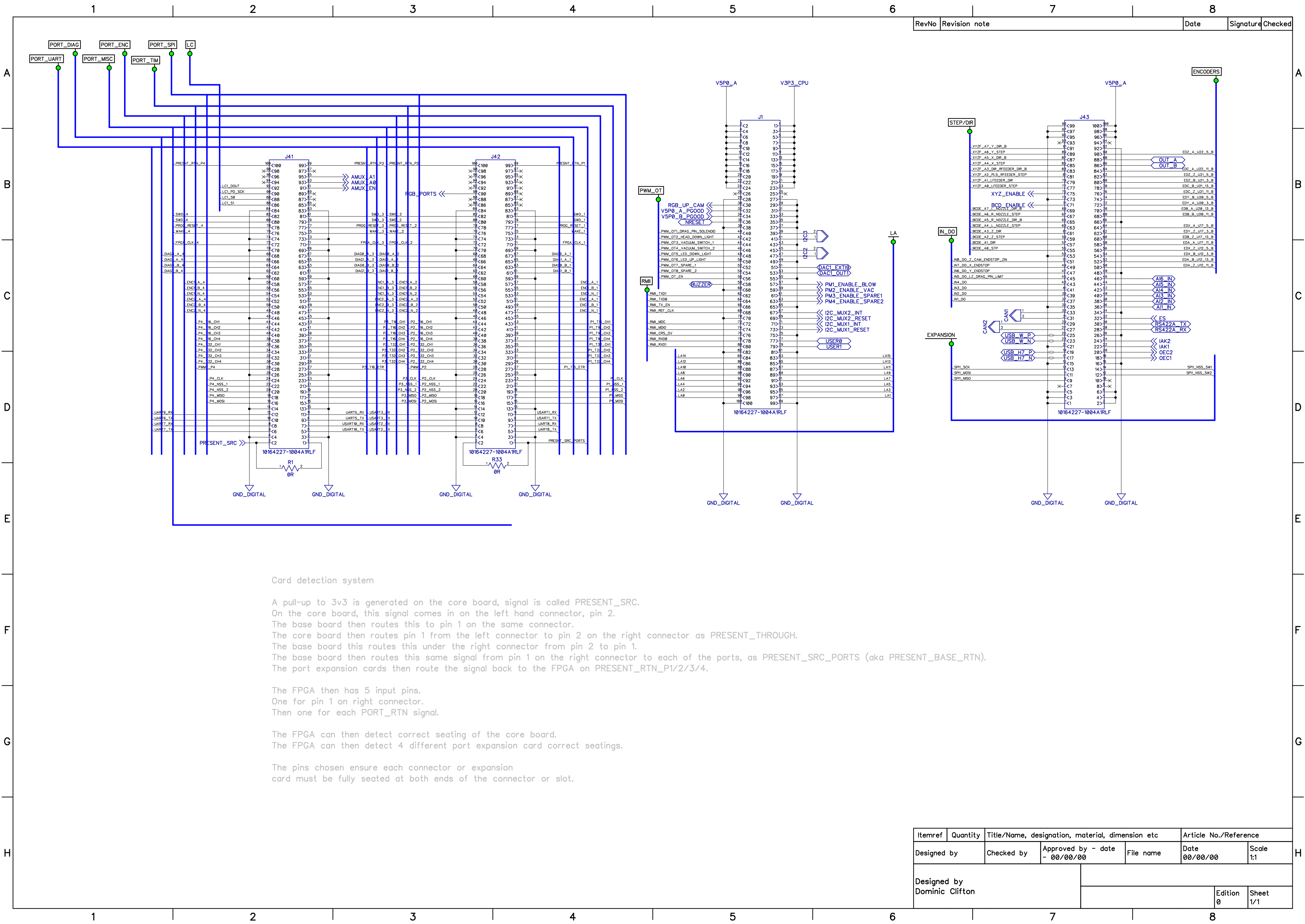
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Card detection system

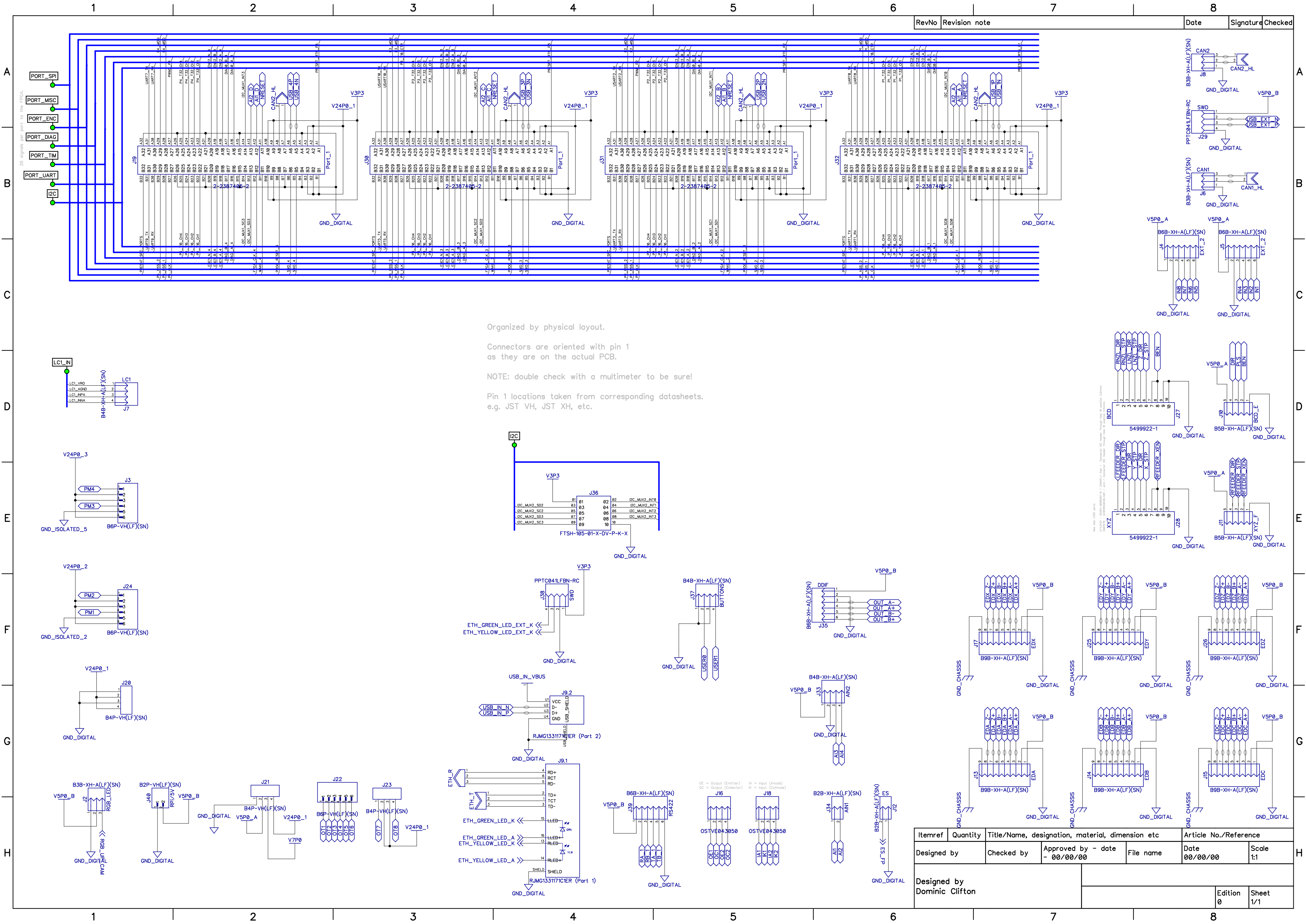
A pull-up to 3v3 is generated on the core board, signal is called PRESENT_SRC.
On the core board, this signal comes in on the left hand connector, pin 2.
The base board then routes this to pin 1 on the same connector.
The core board then routes pin 1 from the left connector to pin 2 on the right connector as PRESENT_THROUGH.
The base board this routes this under the right connector from pin 2 to pin 1.
The base board then routes this same signal from pin 1 on the right connector to each of the ports, as PRESENT_SRC_PORTS (aka PRESENT_BASE_RTN).
The port expansion cards then route the signal back to the FPGA on PRESENT_RTN_P1/2/3/4.

The FPGA then has 5 input pins.
One for pin 1 on right connector.
Then one for each PORT_RTN signal.

The FPGA can then detect correct seating of the core board.
The FPGA can then detect 4 different port expansion card correct seatings.

The pins chosen ensure each connector or expansion card must be fully seated at both ends of the connector or slot.

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Organized by physical layout.

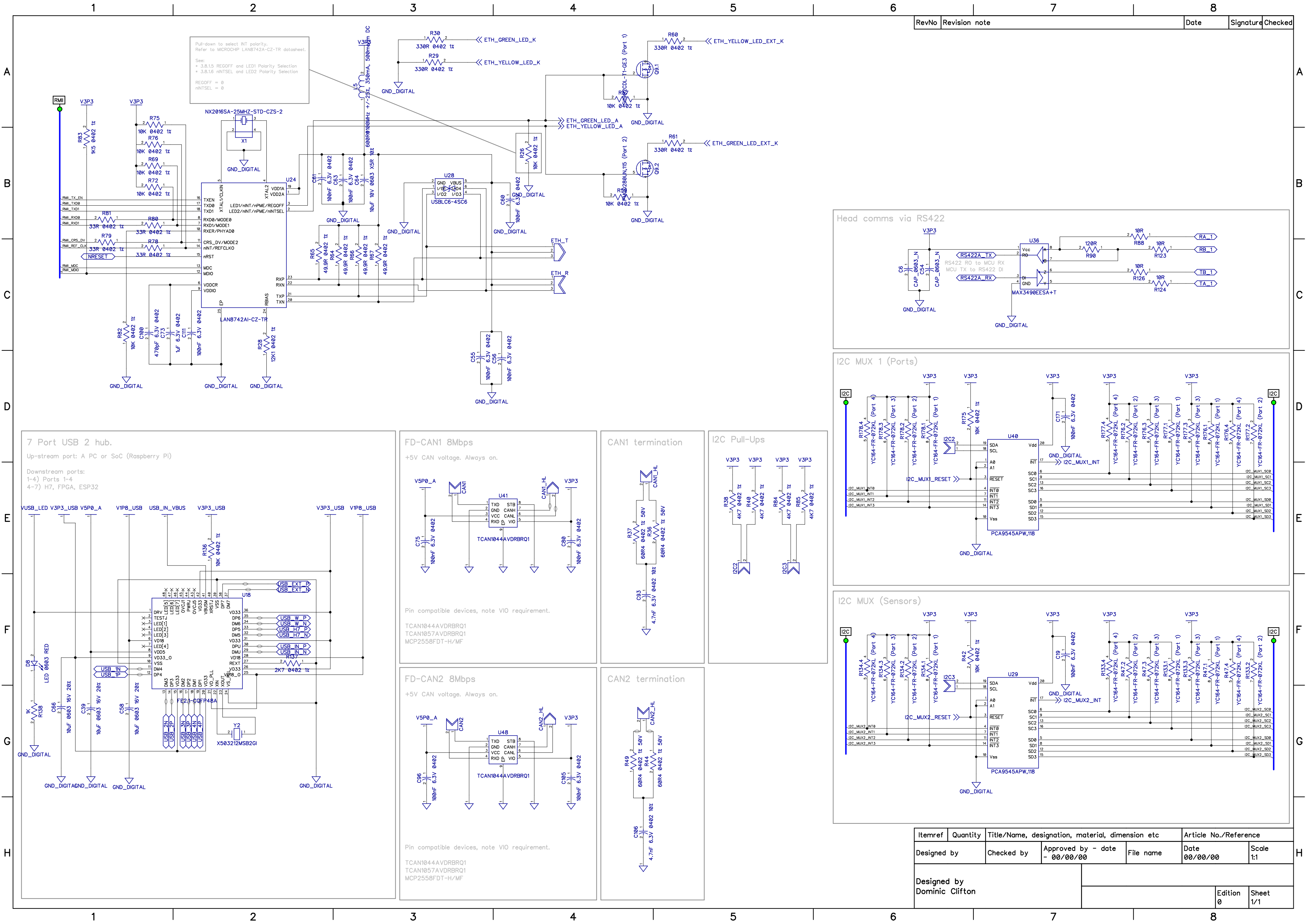
Connectors are oriented with pin 1 as they are on the actual PCB.

NOTE: double check with a multimeter to be sure!

Pin 1 locations taken from corresponding datasheets. e.g. JST VH, JST XH, etc.

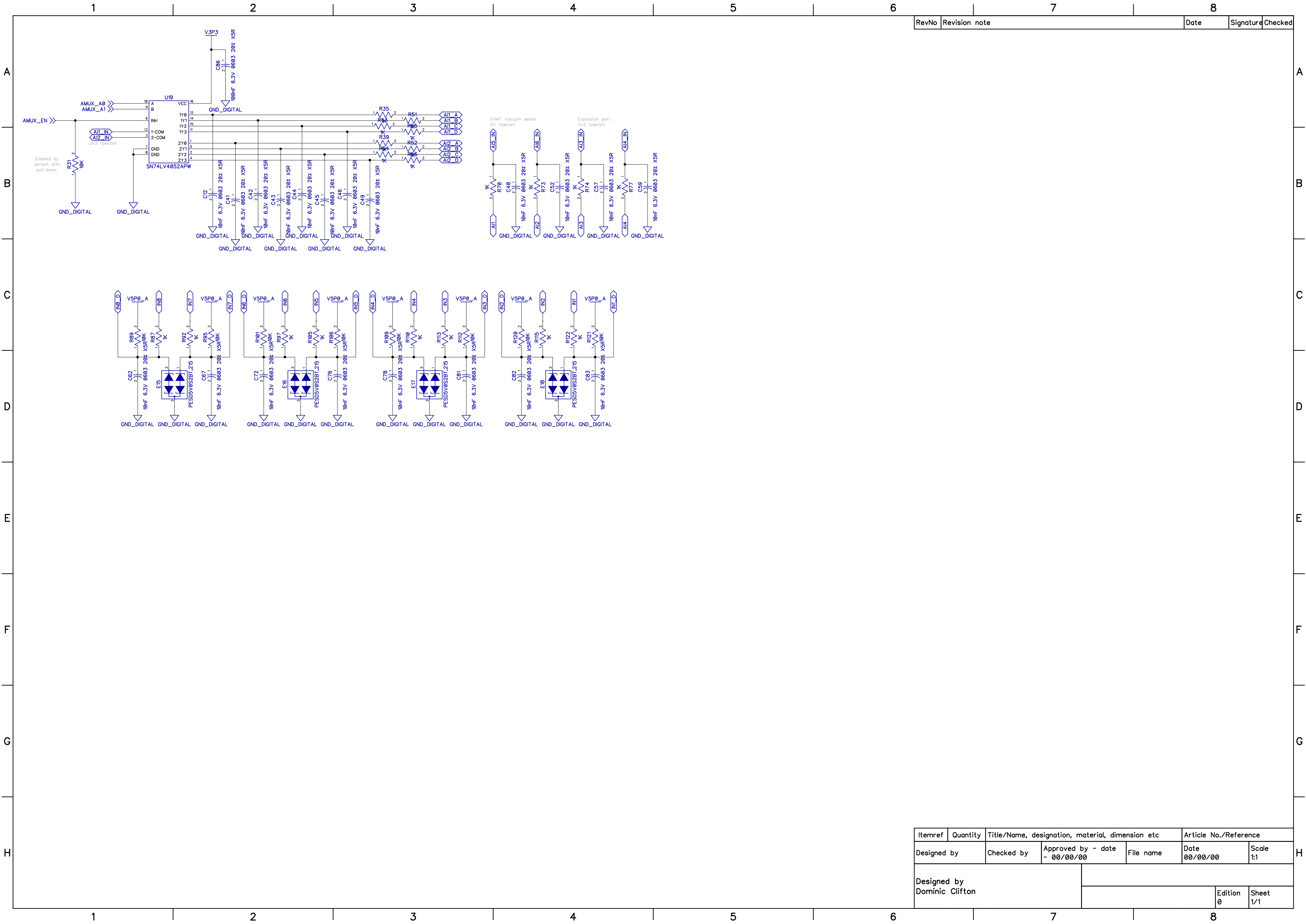
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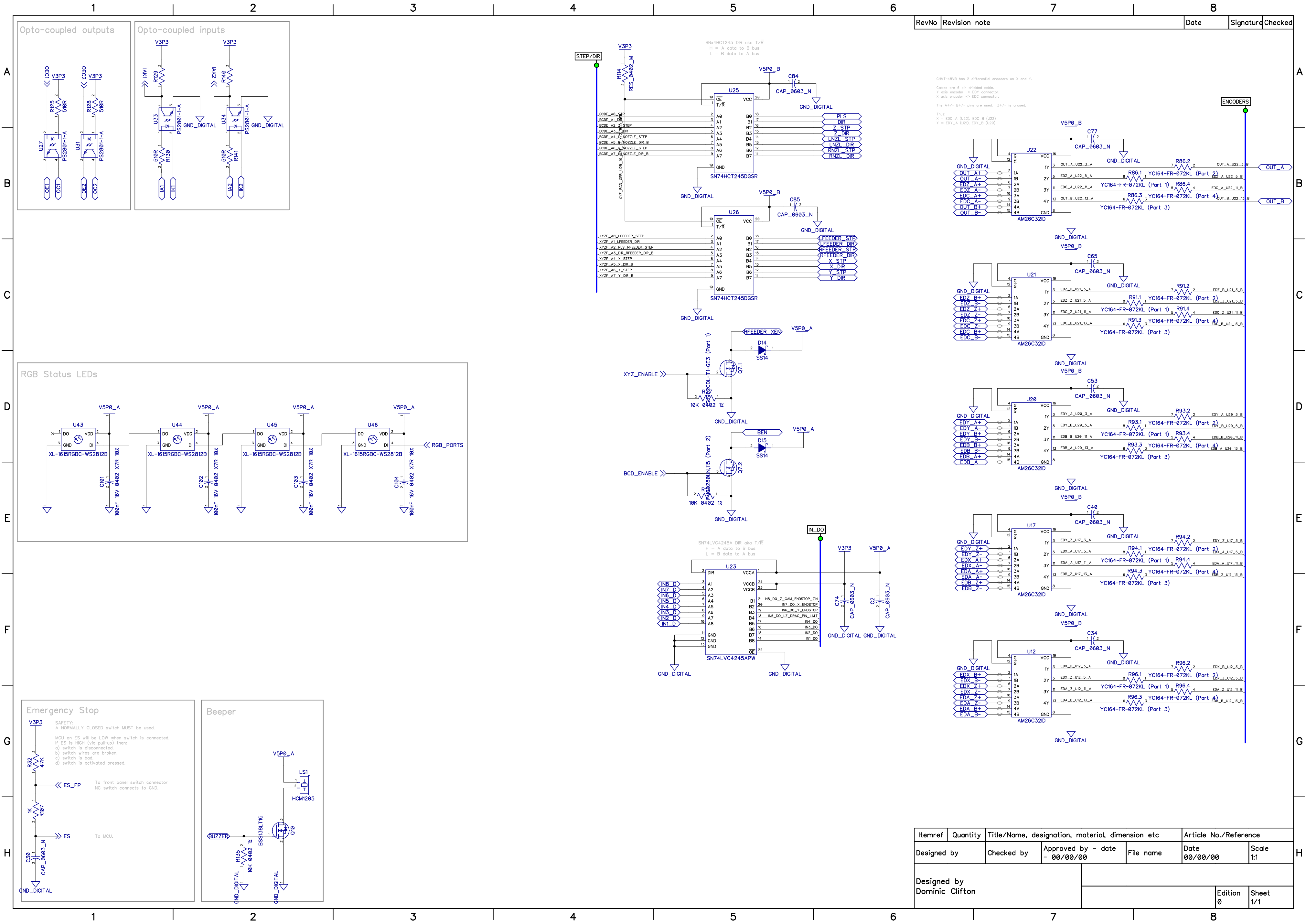
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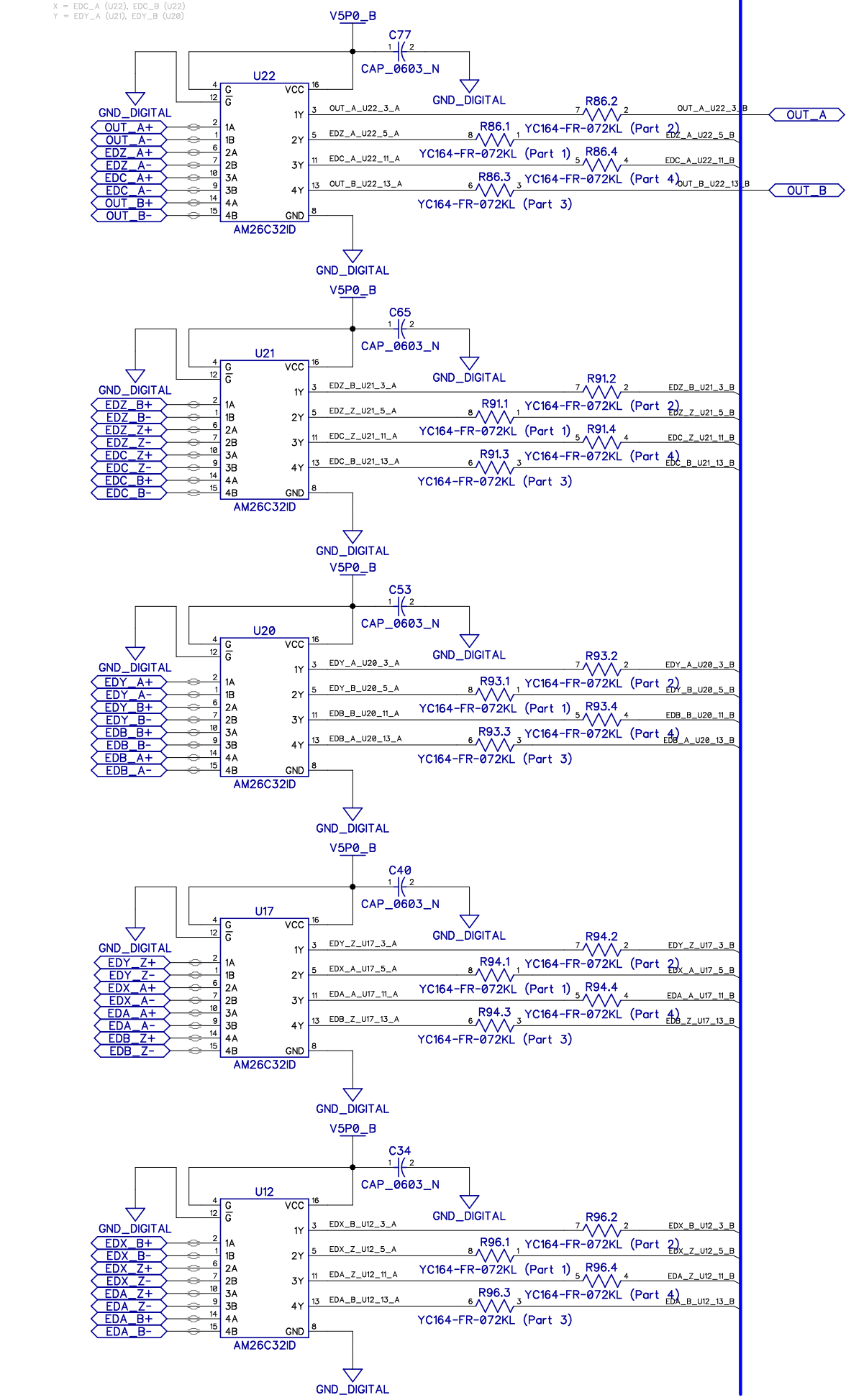
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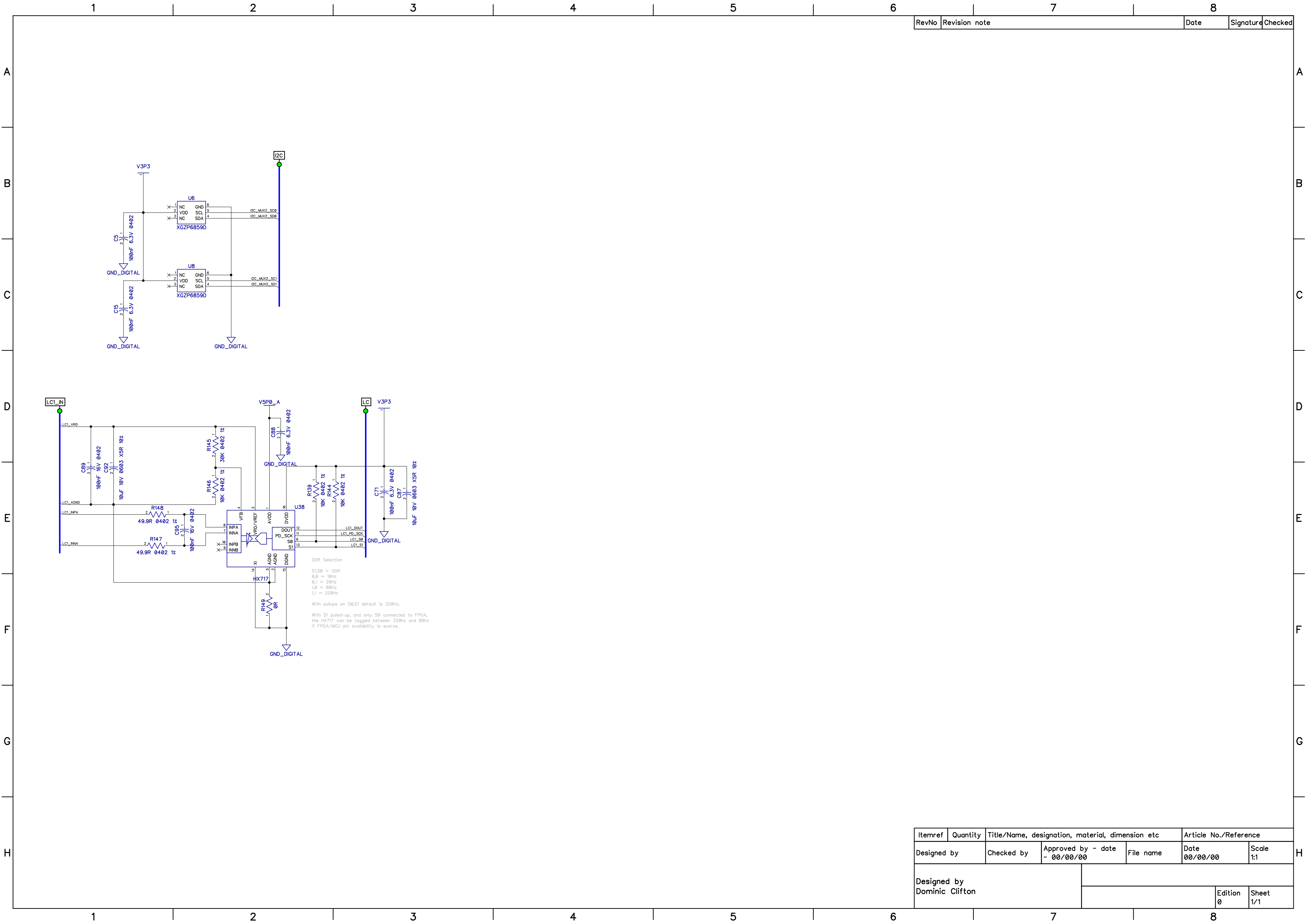


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GM7-48Vb has 2 differential encoders on X and Y.
Cables are 6 pin shielded cables.
Y axis encoder -> EDC connector.
X axis encoder -> EDC connector.
The A+/- B+/- pins are used. Z+/- is unused.
Thus:
X = EDC_A (U22), EDC_B (U23)
Y = EDC_Y (U21), EDC_Z (U26)



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